

A two-point Gibbs family freezes at the minimum

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Conjecture 00000008486

Both source versions specify the Gibbs density $\mu_t \propto e^{-f/t}$ as $t \downarrow 0$, and assert that if the maximum set of f is a periodic orbit, then the limit is the uniform measure on that orbit. The negative sign in the stated density makes this assertion false even for a compact two-point system.

Theorem 1. *There is a continuous function on a compact dynamical system whose maximum set is a fixed orbit, but whose stated Gibbs family converges weakly to a different point mass, rather than the uniform measure on the maximum orbit.*

Proof. Take $\Omega = \{0, 1\}$ with the discrete topology and its Borel sigma-algebra, and let T be the identity. Set $f(0) = 0$ and $f(1) = 1$. The space is compact, both T and f are continuous, and the entire maximum set of f is $\{1\}$, a periodic orbit of period one. Its uniform measure is δ_1 .

Use counting measure on Ω as the reference measure for the Gibbs density. For $t > 0$, its partition function and normalized measure are

$$Z_t = \sum_{x \in \Omega} e^{-f(x)/t} = 1 + e^{-1/t}, \quad \mu_t = \frac{1}{1 + e^{-1/t}} \delta_0 + \frac{e^{-1/t}}{1 + e^{-1/t}} \delta_1.$$

Both weights are positive and sum to one, so this is an actual probability measure with exactly the density prescribed in the source. Using the uniform reference probability instead gives the same normalized Gibbs measure.

As $t \downarrow 0$, the reciprocal $1/t$ tends to infinity, hence $e^{-1/t} \rightarrow 0$. Therefore the weights of 0 and 1 tend respectively to 1 and 0. For every bounded continuous real function g on Ω ,

$$\int g d\mu_t = \frac{g(0)}{1 + e^{-1/t}} + \frac{e^{-1/t}g(1)}{1 + e^{-1/t}} \rightarrow g(0) = \int g d\delta_0.$$

Thus μ_t converges weakly to δ_0 . Since $\delta_0(\{0\}) = 1$ and $\delta_1(\{0\}) = 0$, these two probability measures are distinct. The weak topology on probabilities on this finite discrete space is Hausdorff, so μ_t cannot also converge to δ_1 . This contradicts the asserted maximum-orbit conclusion. $\square$

Scope

The sign is exactly the negative sign appearing in both originals; it is not replaced by the positive sign that would favor maxima. The reference measure gives equal positive weight to both points, and the Gibbs measure is explicitly normalized. No limit point, normalization, or orbit property is assumed. Refuting the maximum-orbit clause suffices to disprove the conjunction, without resolving the other equilibrium or oscillation claims.

Actual probability measures and weak topology in Lean

The state type is `Fin 2` with its discrete topology and actual Borel measurable structure. `compact_space`, `continuous_T` and `continuous_f` prove the topological hypotheses. The definition `maximizers` quantifies over every state; `maximum_set` proves it equals `{1}`. The theorem `maximum_orbit` proves that the range of all actual iterates of T at 1 is precisely this maximum set; `maximum_fixed` proves period one directly.

The nonnegative real weights `p t` and `q t` are defined using the actual real exponential. Their sum is proved to be one. The definition `raw t` is the sum of their scalar multiples of the actual Dirac measures. Its total mass is proved to be one, giving the bundled `ProbabilityMeasure` called `gibbs t`. The theorem `partition_function` evaluates the actual finite exponential sum. For every state, `gibbs_mass` proves exactly

$$\mu_t(\{x\}) = \frac{e^{-f(x)/t}}{\sum_{y \in \Omega} e^{-f(y)/t}}.$$

Point masses specify the whole measure on this finite space, so this is a complete correspondence with the source's Gibbs family relative to counting measure, rather than an asserted probability table.

The theorem `integral_gibbs` computes genuine Bochner integrals against the constructed measure. `weight_limit`, `p_limit` and `q_limit` prove the exponential and normalized weight limits. The theorem `weak_limit_minimum` then proves convergence in Mathlib's actual weak topology on `ProbabilityMeasure`, using its characterization by integrals against every bounded continuous real function. No custom convergence relation is substituted.

The definitions `point 0` and `point 1` are actual Dirac probability measures.

`point_masses_distinct` distinguishes them by their measures of `{0}`. Finally,

`not_weak_limit_maximum` negates actual weak convergence to `point 1`, using uniqueness of limits in the Hausdorff probability-measure topology.

All convergence statements use the actual right-hand neighborhood filter `nhdsWithin 0 (Set.Ioi 0)`. Totalized real division lets the definitions also exist at temperature zero, but that value plays no role in these positive-temperature limits.

Reproduction

Lean 4.19.0 and Mathlib are pinned at commit

```
c44e0c8ee63ca166450922a373c7409c5d26b00b.
```

From `lean/`, run `lake update`, `lake exe cache get`, and `lake build`; then check the complete source with

```
lake env lean Main.lean -DwarningAsError=true.
```

The source prints five principal dependency audits. Only standard logical axioms are used, with no admitted statements or auxiliary computations.